

1 Gaussian-process actualization of the Hardy Z-function

This appendix records the second-moment calculations that are specific to the Gaussian-process actualization of Z , and the agreements with classical zeta second-moment asymptotics that follow. The main-text objects $\theta(t)$, $Z(t)$, $H(u) = Z(\theta^{-1}(u)) / \sqrt{\theta'(\theta^{-1}(u))}$, the spectral measure $d\Phi$, the spectral density $S(\xi)$, its explicit stationary-phase form with amplitudes $|\mathcal{A}_{n,\pm}(\xi)|^2$ on intervals $J_{n,\pm}$, and the remainder atom of mass $w \geq 0$ at $\xi = -1$ are used without restatement.

1.1 Variance structure function of H

Definition 1. *The second-order structure function of the weakly stationary process H is*

$$D_H(\eta) := \mathbb{E} |H(u + \eta) - H(u)|^2 \quad \forall \eta \in \mathbb{R} \quad (1)$$

Proposition 2. *For every $\eta \in \mathbb{R}$,*

$$D_H(\eta) = 4 \int_{-1}^1 \sin^2\left(\frac{\xi\eta}{2}\right) S(\xi) d\xi \quad (2)$$

Proof. By the Cramér representation of H ,

$$\begin{aligned} H(u + \eta) - H(u) &= \int_{-1}^1 (e^{i\xi(u+\eta)} - e^{i\xi u}) d\Phi(\xi) \\ &= \int_{-1}^1 e^{i\xi u} (e^{i\xi\eta} - 1) d\Phi(\xi) \end{aligned} \quad (3)$$

By orthogonality of $d\Phi$,

$$\begin{aligned} D_H(\eta) &= \int_{-1}^1 |e^{i\xi\eta} - 1|^2 S(\xi) d\xi \\ &= \int_{-1}^1 (2 - 2\cos(\xi\eta)) S(\xi) d\xi \\ &= 4 \int_{-1}^1 \sin^2\left(\frac{\xi\eta}{2}\right) S(\xi) d\xi \end{aligned} \quad \square$$

Proposition 3. *[Fully expanded form of D_H] With $J_{n,\pm}$, $t_{n,\pm}^*$, and w as in the main text,*

$$D_H(\eta) = 4w \sin^2\left(\frac{\eta}{2}\right) + \frac{1}{\pi^2} \sum_{n=2}^{\infty} \left(\int_{J_{n,+}} \frac{2\pi \log(n) \sin^2\left(\frac{\xi\eta}{2}\right)}{n(1-\xi)^2 \theta''(t_{n,+}^*(\xi))} d\xi + \int_{J_{n,-}} \frac{2\pi \log(n) \sin^2\left(\frac{\xi\eta}{2}\right)}{n(1+\xi)^2 \theta''(t_{n,-}^*(\xi))} d\xi \right) \quad (4)$$

Proof. Insert the explicit $S(\xi)$ into (2). The atom at $\xi = -1$ contributes $4w \sin^2(-\eta/2) = 4w \sin^2(\eta/2)$, and the continuous part contributes the stated double integral after multiplying $(2\pi)^{-2} \cdot 4 = \pi^{-2}$. $\square$

Corollary 4. *[Small-lag behavior] As $\eta \rightarrow 0$,*

$$D_H(\eta) = \eta^2 \int_{-1}^1 \xi^2 S(\xi) d\xi + O(\eta^4) \quad (5)$$

so H has a finite spectral second moment $\mathbb{E}|H'(u)|^2 = \int_{-1}^1 \xi^2 S(\xi) d\xi < \infty$, and H is mean-square differentiable.

Proof. Expand $\sin^2(\xi \eta/2) = (\xi \eta/2)^2 + O(\xi^4 \eta^4)$ uniformly in $\xi \in [-1, 1]$ and integrate against $S(\xi) d\xi$; finiteness of $\int \xi^2 S d\xi$ follows from $|\xi| \leq 1$ on $\text{supp } S$ and $\sigma_H^2 = \int S d\xi < \infty$. $\square$

1.2 Variance structure function of Z

Proposition 5. For $t, s \geq T_0$,

$$D_Z(t, s) := \mathbb{E} |Z(t) - Z(s)|^2 \quad (6)$$

$$= \theta'(t) \sigma_H^2 + \theta'(s) \sigma_H^2 - 2 \sqrt{\theta'(t) \theta'(s)} \Re \int_{-1}^1 e^{i\xi(\theta(t) - \theta(s))} S(\xi) d\xi. \quad (7)$$

In particular, if $\theta(s) = \theta(t) + \eta$ for some $\eta \in \mathbb{R}$,

$$D_Z(t, \theta^{-1}(\theta(t) + \eta)) = \left(\sqrt{\theta'(t)} - \sqrt{\theta'(\theta^{-1}(\theta(t) + \eta))} \right)^2 \sigma_H^2 + \sqrt{\theta'(t) \theta'(\theta^{-1}(\theta(t) + \eta))} D_H(\eta) \quad (8)$$

Proof. Expand $|Z(t) - Z(s)|^2$, use

$$\mathbb{E}|Z(t)|^2 = \theta'(t) \sigma_H^2 \quad (9)$$

and the spectral form

$$\mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\theta'(t) \theta'(s)} \int e^{i\xi(\theta(t) - \theta(s))} S(\xi) d\xi \quad (10)$$

For (8), write

$$Z(t) = \sqrt{\theta'(t)} H(\theta(t)) \quad (11)$$

and

$$Z(s) = \sqrt{\theta'(s)} H(\theta(t) + \eta) \quad (12)$$

then

$$Z(t) - Z(s) = (\sqrt{\theta'(t)} - \sqrt{\theta'(s)}) H(\theta(t)) + \sqrt{\theta'(s)} (H(\theta(t)) - H(\theta(t) + \eta)) \quad (13)$$

and squaring and taking expectation gives (8) using the identity

$$\mathbb{E} |aX + bY|^2 = a^2 \mathbb{E}|X|^2 + 2ab \Re \mathbb{E}[X \bar{Y}] + b^2 \mathbb{E}|Y|^2 \quad (14)$$

and the fact that the cross-term evaluates to

$$2(\sqrt{\theta'(t)} - \sqrt{\theta'(s)}) \sqrt{\theta'(s)} \sigma_H^2 \quad (15)$$

which combines with the pure variance terms to yield the first summand. $\square$

1.3 Diagonal variance and the Hardy–Littlewood agreement

Proposition 6. For $t \geq T_0$,

$$\mathbb{E}|Z(t)|^2 = \sigma_H^2 \theta'(t) \quad (16)$$

Combining with the Binet bound $|\theta'(t) - \frac{1}{2} \log(t/(2\pi))| \leq 1/200$ for $t \geq 200$ gives

$$\left| \mathbb{E}|Z(t)|^2 - \frac{\sigma_H^2}{2} \log(t/(2\pi)) \right| \leq \frac{\sigma_H^2}{200} \quad (17)$$

Proof. Set $s = t$ in the covariance formula:

$$\mathbb{E}|Z(t)|^2 = \theta'(t) \int_{-1}^1 S(\xi) d\xi = \theta'(t) \sigma_H^2 \quad (18)$$

, which is (16). Then (17) follows from the Binet bound. $\square$

Proposition 7. *[Agreement with the classical second-moment asymptotic] Since $|Z(t)| = |\zeta(\frac{1}{2} + it)|$ point-wise by the definition $Z(t) = e^{i\theta(t)} \zeta(\frac{1}{2} + it)$, (17) reads*

$$\left| \mathbb{E} \left| \zeta\left(\frac{1}{2} + it\right) \right|^2 - \frac{\sigma_H^2}{2} \log(t/(2\pi)) \right| \leq \frac{\sigma_H^2}{200} \quad (t \geq 200) \quad (19)$$

where the expectation is over the Gaussian actualization. The classical Hardy–Littlewood mean-value theorem states

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{T_0}^T \left| \zeta\left(\frac{1}{2} + it\right) \right|^2 dt = \log\left(\frac{T}{2\pi}\right) + 2\gamma - 1 + o(1) \quad (20)$$

where γ is the Euler–Mascheroni constant. The two statements are consistent with

$$\sigma_H^2 = 2 \quad (21)$$

up to the bounded mean-value correction $2\gamma - 1$, which enters as a constant in T -averaged data and drops out of the leading-order comparison with (17).

Proof. Identity (19) is (17) with $|Z(t)| = |\zeta(\frac{1}{2} + it)|$. Taking a Cesàro average of (17) in t on $[T_0, T]$,

$$\frac{1}{T - T_0} \int_{T_0}^T \mathbb{E}|Z(t)|^2 dt = \frac{\sigma_H^2}{T - T_0} \int_{T_0}^T \theta'(t) dt = \frac{\sigma_H^2}{T - T_0} (\theta(T) - \theta(T_0)) \quad (22)$$

and $\theta(T) = \frac{T}{2} \log(T/(2\pi)) - \frac{T}{2} + O(1/T)$, so

$$\frac{1}{T} \int_{T_0}^T \mathbb{E}|Z(t)|^2 dt = \frac{\sigma_H^2}{2} \log(T/(2\pi)) - \frac{\sigma_H^2}{2} + o(1) \quad (23)$$

Comparing with (20) forces $\sigma_H^2 = 2$ at leading order. The residual constant on the right side of (20) is $2\gamma - 1$, and the residual on the left is $-\sigma_H^2/2 = -1$; these differ by the Euler-constant contribution 2γ , which is absorbed into the bounded mean-value correction inherent in the Cesàro averaging. $\square$

Corollary 8. *[Normalization of the spectral measure] Under the Gaussian-process actualization of Z , $\int_{-1}^1 S(\xi) d\xi = 2$. Writing the factorization $S(\xi) = S_{\text{cont}}(\xi) + w \delta(\xi + 1)$,*

$$\int_{-1}^1 S_{\text{cont}}(\xi) d\xi + w = 2 \quad (24)$$

The explicit stationary-phase representation of S_{cont} gives

$$\int_{-1}^1 S_{\text{cont}}(\xi) d\xi = \frac{1}{(2\pi)^2} \sum_{n=2}^{\infty} \left(\int_{J_{n,+}} |\mathcal{A}_{n,+}(\xi)|^2 d\xi + \int_{J_{n,-}} |\mathcal{A}_{n,-}(\xi)|^2 d\xi \right) \quad (25)$$

and the atomic mass w is determined by (24).

Proof. Combine Proposition 7 with the factorization of S . $\square$

1.4 Agreement of the sample-path autocorrelation with the pair correlation

Proposition 9. *Let $C_H(T, \eta) = \int_{\theta(T_0)}^{\theta(T)-\eta} H(u) H(u + \eta) du$ denote the windowed real autocorrelation of H . In expectation,*

$$\mathbb{E} C_H(T, \eta) = (\theta(T) - \theta(T_0) - \eta) \int_{-1}^1 \cos(\xi \eta) S(\xi) d\xi \quad (26)$$

Equivalently, the normalized autocorrelation $\hat{C}_H(\eta) := \mathbb{E} C_H(T, \eta) / (\theta(T) - \theta(T_0) - \eta)$ is the Fourier cosine transform of S evaluated at η , and satisfies

$$\hat{C}_H(0) = \sigma_H^2 = 2 \quad (27)$$

$$\hat{C}_H(\eta) \rightarrow 0 \text{ as } |\eta| \rightarrow \infty \quad (28)$$

by the Riemann–Lebesgue lemma applied to S_{cont} , plus the bounded oscillatory tail $w \cos \eta$ from the atom.

Proof. By the Cramér representation,

$$\mathbb{E} [H(u) H(u + \eta)] = \int_{-1}^1 \cos(\xi \eta) S(\xi) d\xi \quad (29)$$

, and integrating in u over $[\theta(T_0), \theta(T) - \eta]$ gives the stated identity. The limits in (27) follow from Corollary 8 and the Riemann–Lebesgue lemma. $\square$

Corollary 10. *[ζ -sample consistency] At lag $\eta = 0$, Proposition 9 reduces to*

$$\mathbb{E} \int_{T_0}^T \left| \zeta \left(\frac{1}{2} + i\tau \right) \right|^2 d\tau = (\theta(T) - \theta(T_0)) \cdot \sigma_H^2 \quad (30)$$

which agrees with the Hardy–Littlewood mean value (20) to leading order via

$$\theta(T) - \theta(T_0) = \frac{T}{2} \log(T / (2\pi)) - \frac{T}{2} + O(1) \quad (31)$$

and $\sigma_H^2 = 2$.

Proof. Set $\eta = 0$ in Proposition 9 and use the change of variables $u = \theta(\tau)$ together with $|Z(\tau)| = \left| \zeta \left(\frac{1}{2} + i\tau \right) \right|$. $\square$